



Introduction to Linux for DevOps

Muhammad Ali

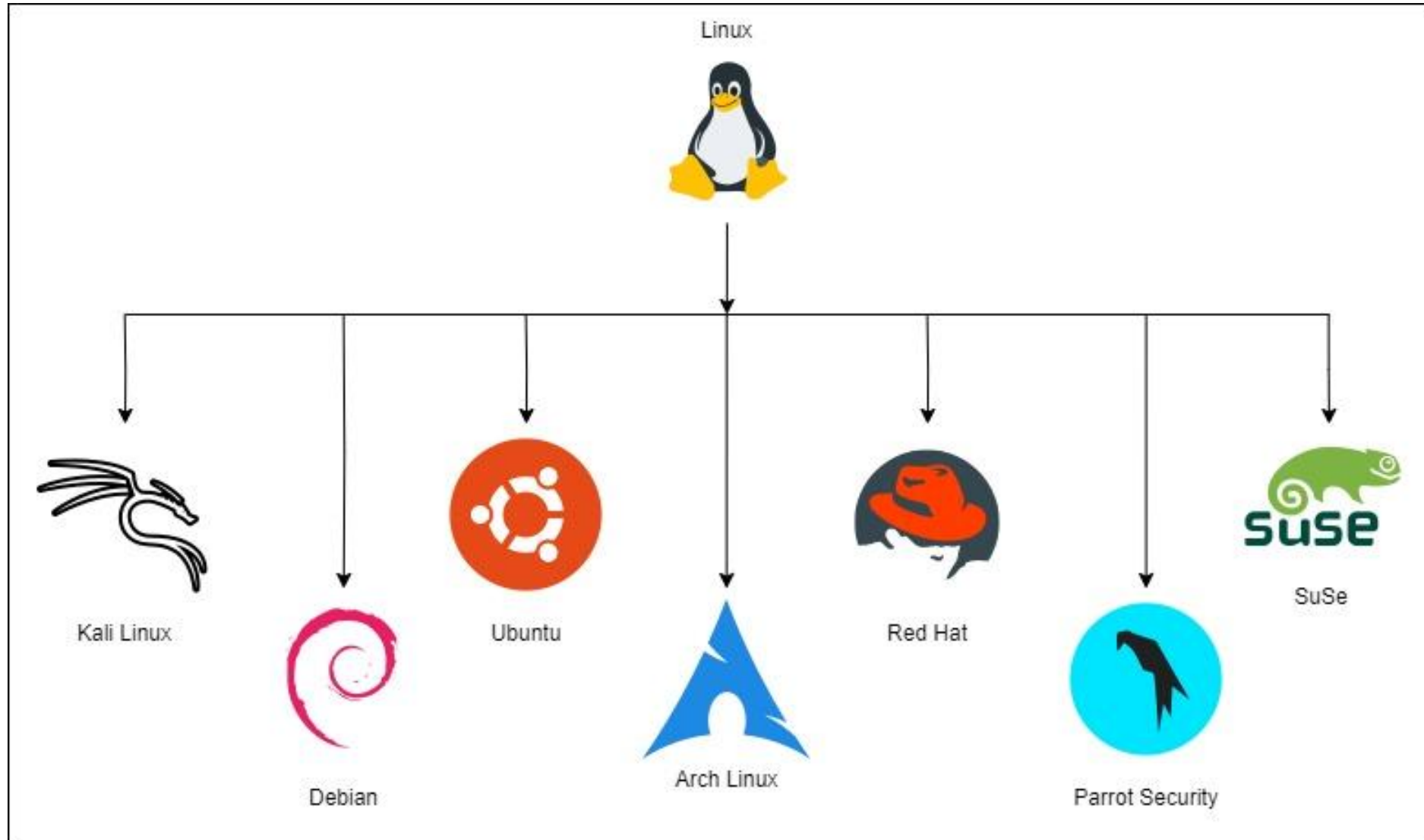
Course : DevOps and Cloud Computing Foundations
Department of Computer Science, SCIT, Beaconhouse National University
(BNU)

What is Linux?

- Linux is an open-source, Unix-like operating system.
- It was created by **Linus Torvalds** in 1991.
- Linux is widely used in servers, desktops, and embedded systems.
- Linux is **free** and **open-source** under the GNU General Public License (GPL), which means anyone can modify and distribute it.

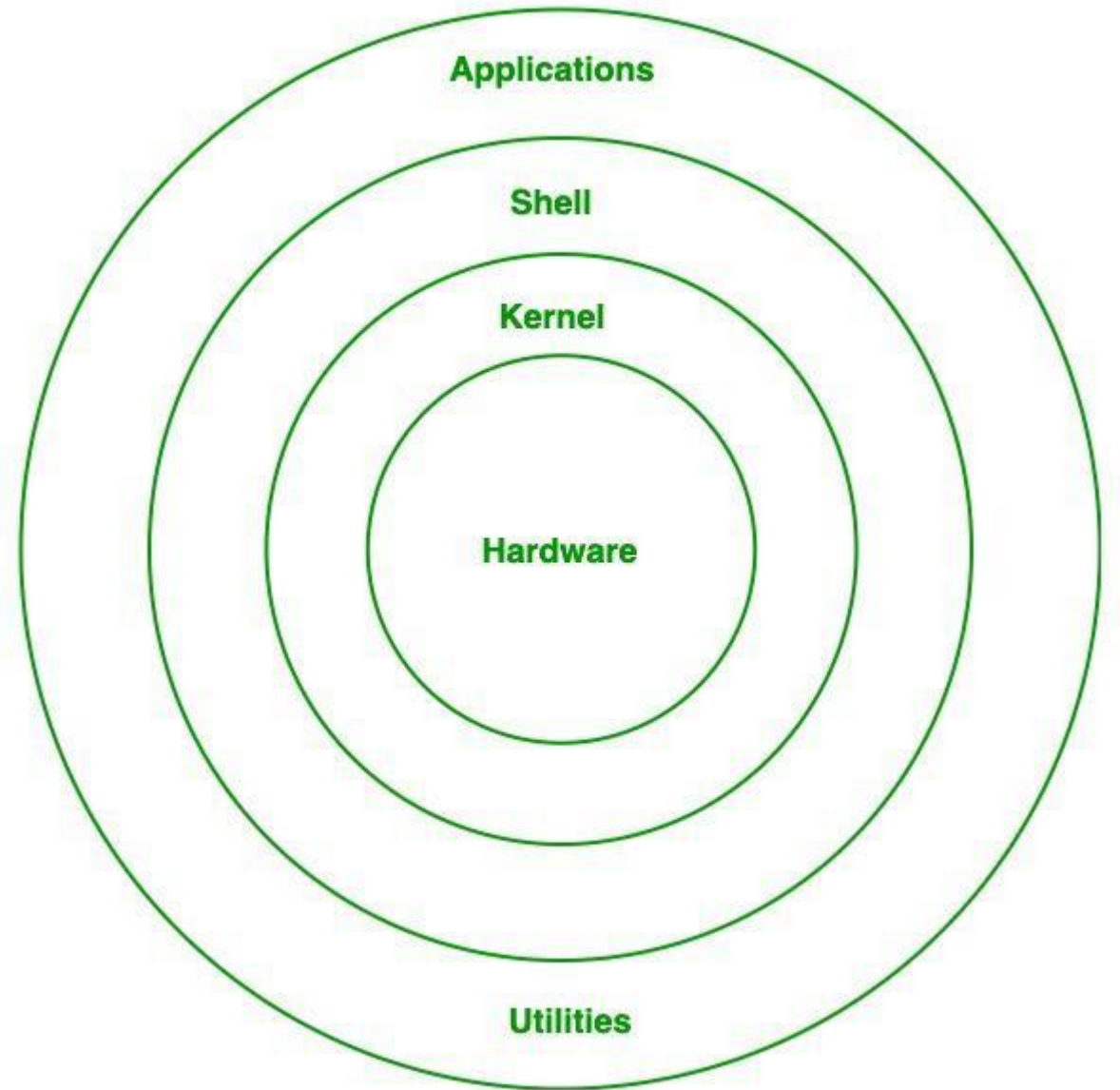


Linux distros



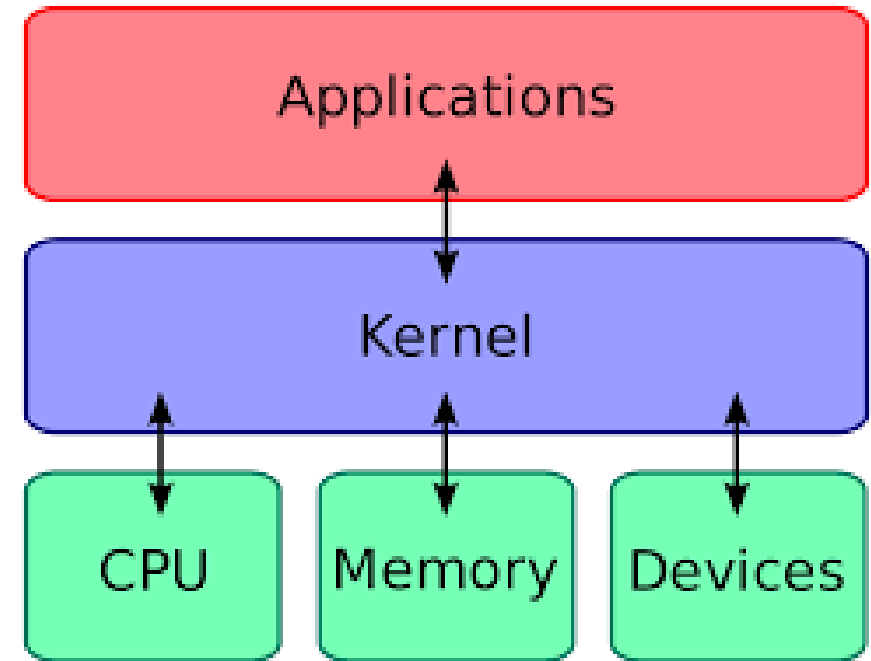
The Linux Architecture

- Kernal
- Shell
- File System
- System Utilities



Kernel

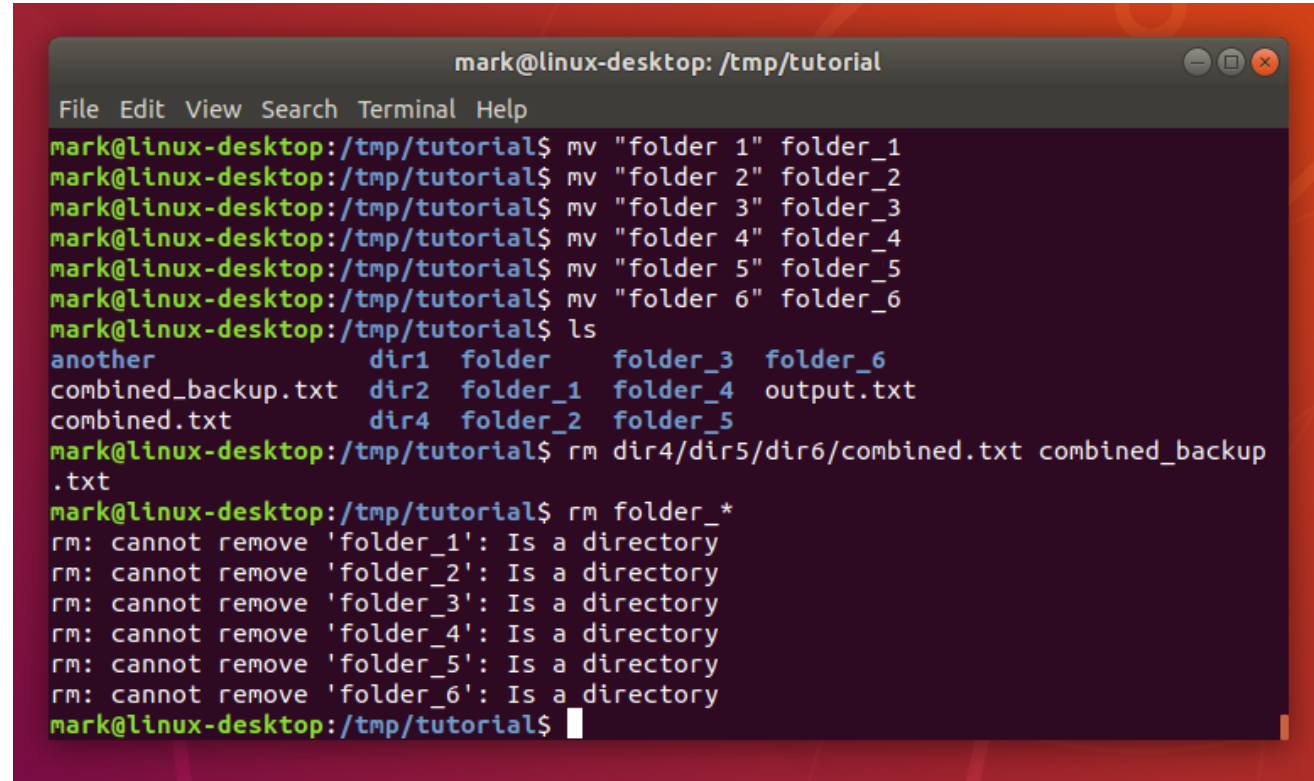
- The kernel is the core of the Linux operating system that manages hardware resources and controls communication between software and hardware.
- Handles process management, memory, and device control.
- Prevents conflicts between multiple running programs.



Shell

- Interface between user and Kernel
- Types of shell, Bash, Zsh, Fish, etc
- The interprets commands and passes it to kernel

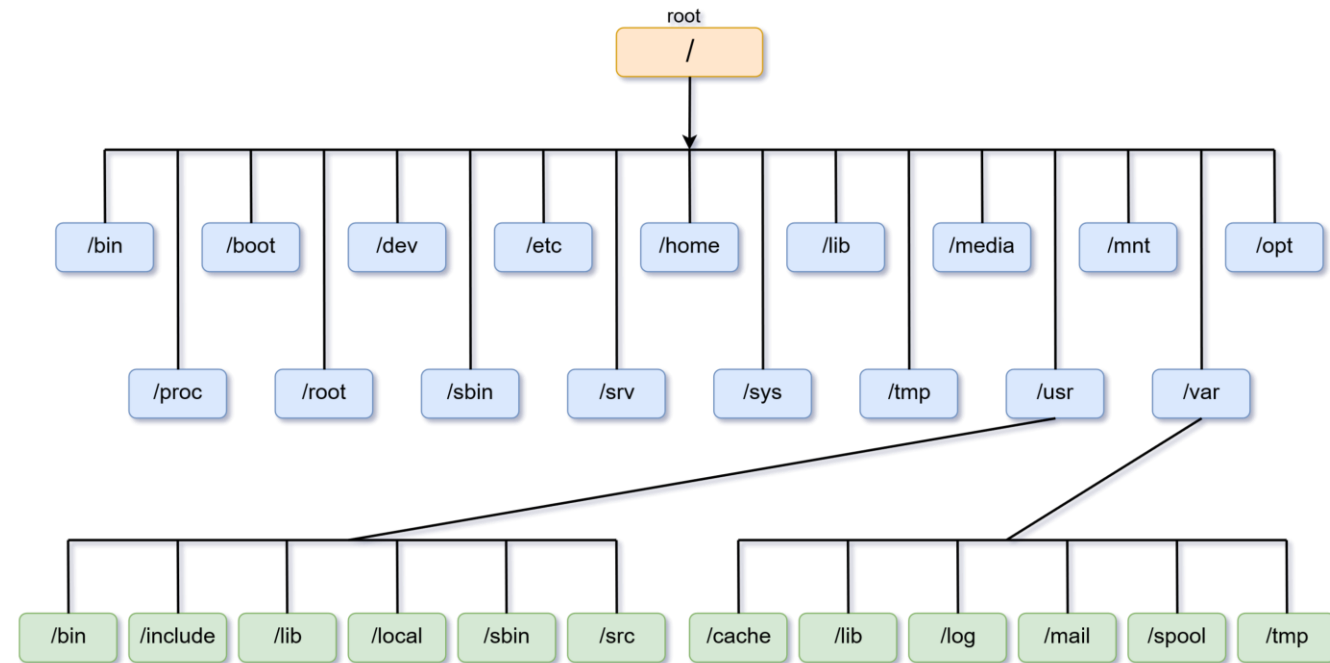
“The **shell** is the software that interprets and executes the various commands that we type in the terminal.”

A terminal window titled 'mark@linux-desktop: /tmp/tutorial' with a menu bar (File, Edit, View, Search, Terminal, Help). The terminal shows a series of commands and their outputs. The user moves six folders to new names, lists the directory contents, and then attempts to remove them. The removal of directories fails with the message 'rm: cannot remove 'folder_X': Is a directory' for each folder.

```
mark@linux-desktop: /tmp/tutorial
File Edit View Search Terminal Help
mark@linux-desktop:/tmp/tutorial$ mv "folder 1" folder_1
mark@linux-desktop:/tmp/tutorial$ mv "folder 2" folder_2
mark@linux-desktop:/tmp/tutorial$ mv "folder 3" folder_3
mark@linux-desktop:/tmp/tutorial$ mv "folder 4" folder_4
mark@linux-desktop:/tmp/tutorial$ mv "folder 5" folder_5
mark@linux-desktop:/tmp/tutorial$ mv "folder 6" folder_6
mark@linux-desktop:/tmp/tutorial$ ls
another          dir1  folder  folder_3  folder_6
combined_backup.txt  dir2  folder_1 folder_4  output.txt
combined.txt       dir4  folder_2 folder_5
mark@linux-desktop:/tmp/tutorial$ rm dir4/dir5/dir6/combined.txt combined_backup
.txt
mark@linux-desktop:/tmp/tutorial$ rm folder_*
rm: cannot remove 'folder_1': Is a directory
rm: cannot remove 'folder_2': Is a directory
rm: cannot remove 'folder_3': Is a directory
rm: cannot remove 'folder_4': Is a directory
rm: cannot remove 'folder_5': Is a directory
rm: cannot remove 'folder_6': Is a directory
mark@linux-desktop:/tmp/tutorial$
```

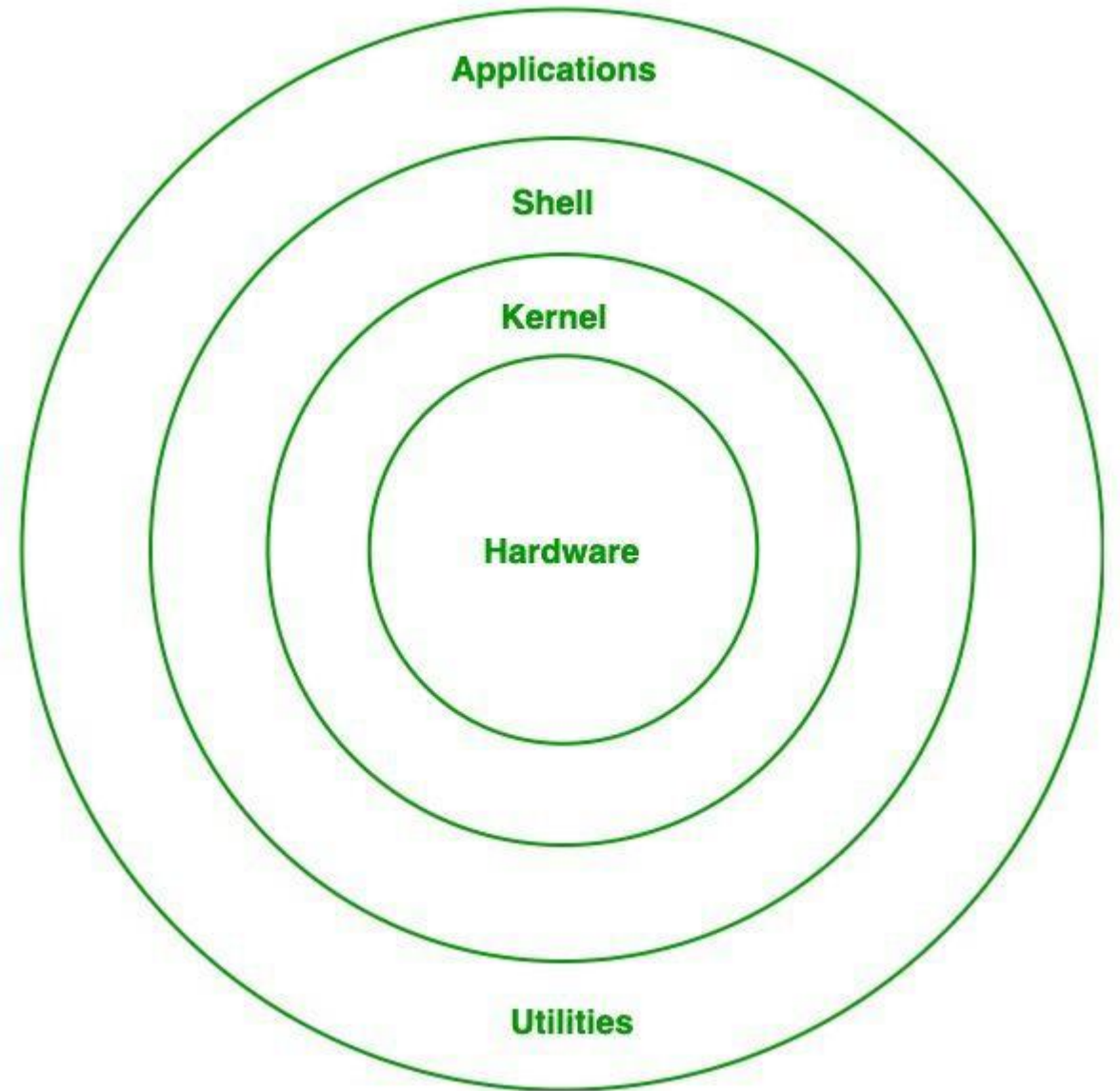
File System

- Linux uses a hierarchical file system. Everything is a file (even devices and processes).
- Key directories:
 - /bin – essential system binaries
 - /etc – system configuration files
 - /home – user home directories
 - /root – root user's home directory
 - /var – variable data (logs, databases)
 - /dev – device files (hard drives, printers)
 - /lib – shared libraries



System Utilities

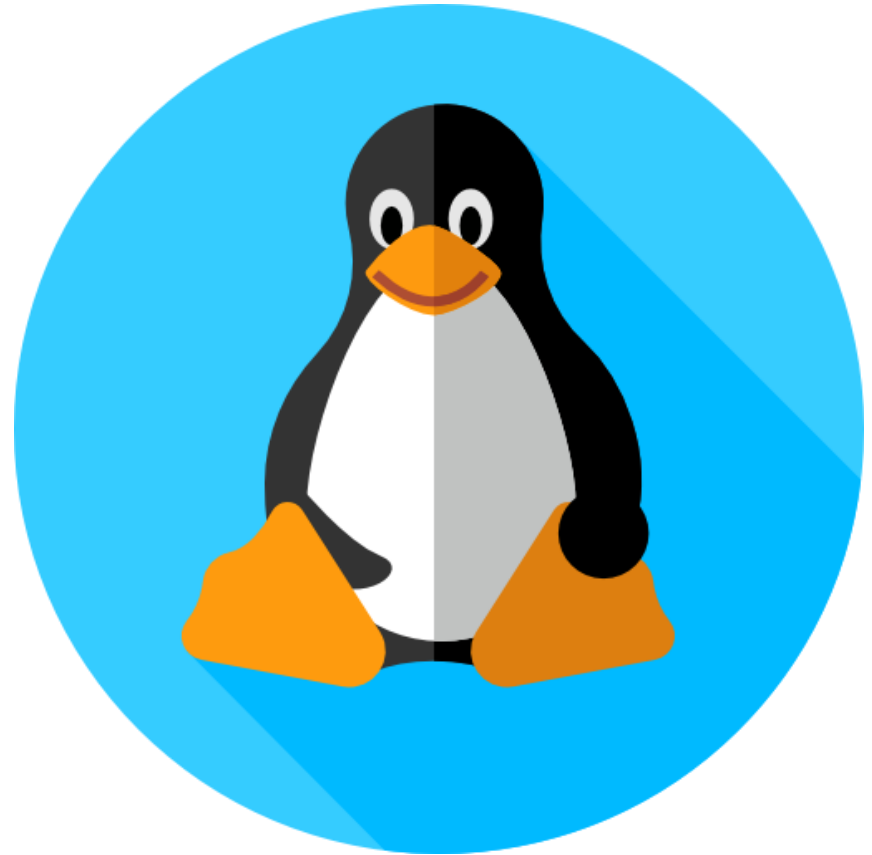
- System utilities are built-in tools that help users manage, configure, and maintain the operating system.
- Used for tasks like software installation, user management, and monitoring.
- Simplify system administration processes for both beginners and administrators.



Basic Linux Commands

Navigating the File System

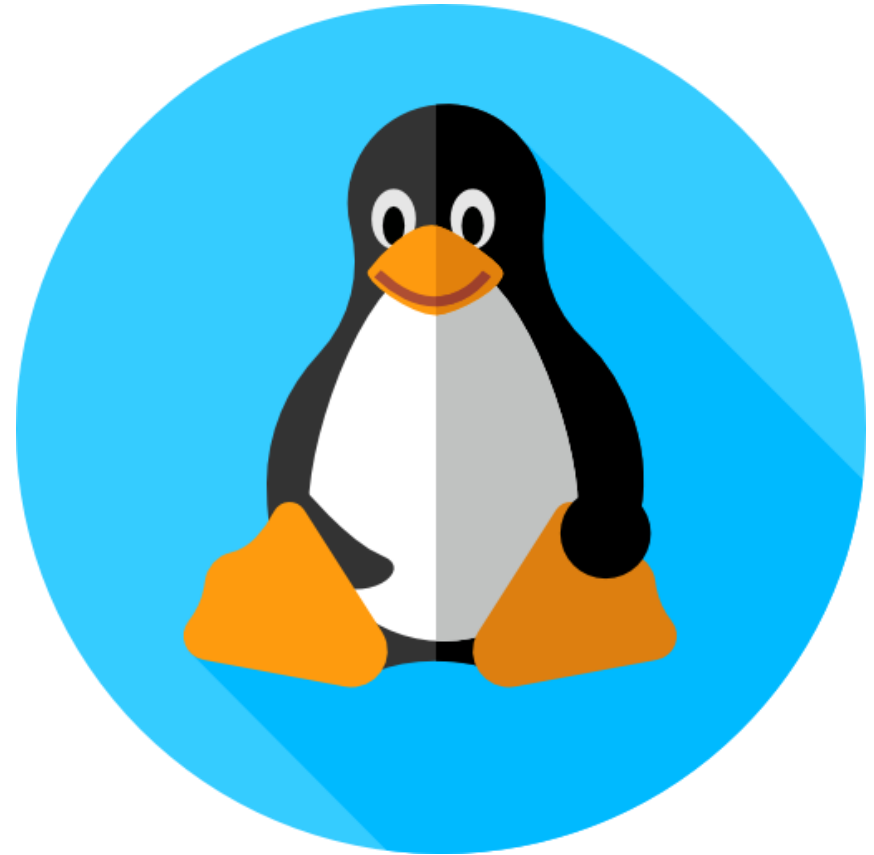
- pwd (print working directory)
- ls (list files and directories)
- cd (change directory)
- mkdir (make directory)
- rmdir (remove directory)
- ls -l (long listing)
- cd ~ (go to home directory)



Basic Linux Commands

File Operations

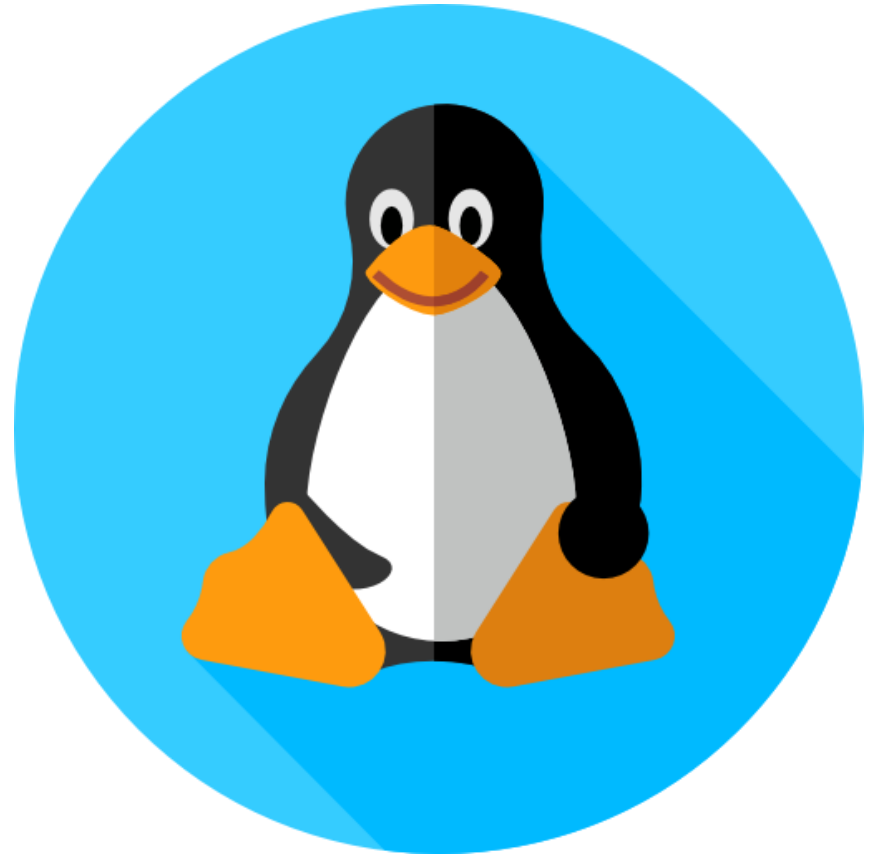
- `touch` (create an empty file)
- `cat` (view file content)
- `cp` (copy files)
- `mv` (move files)
- `rm` (remove files)
- `nano` or `vim` (edit files)



Basic Linux Commands

Managing Processes

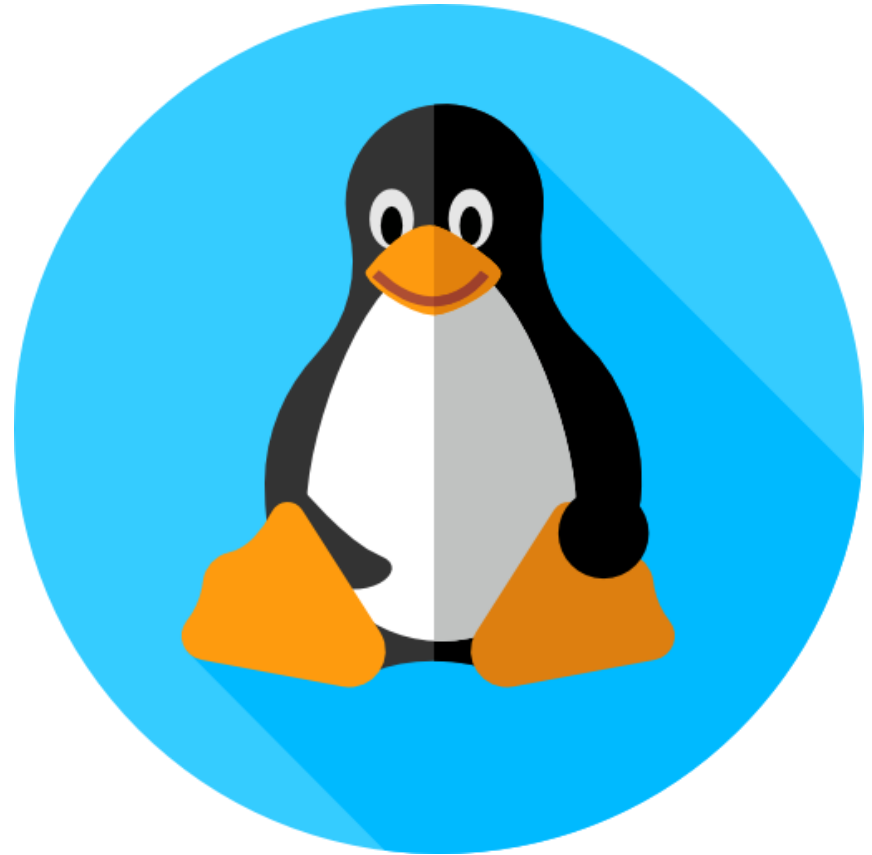
- `ps` (view current processes)
- `top` (real-time process monitor)
- `kill` (terminate a process)



Basic Linux Commands

User and Group Management

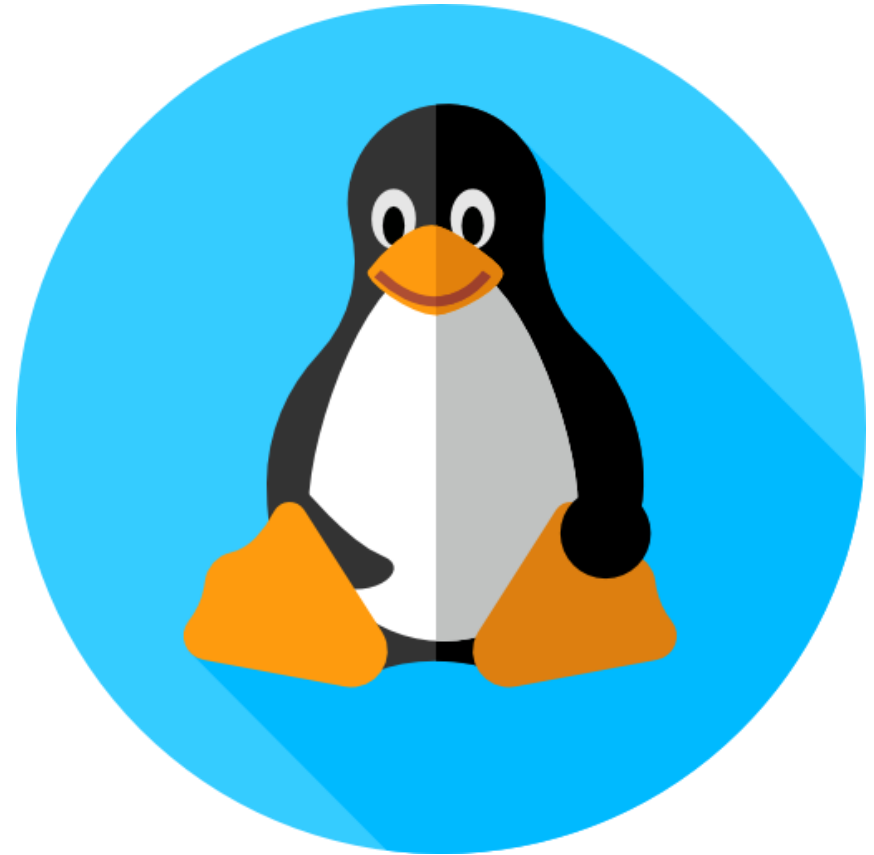
- `whoami` (show current user)
- `id` (show user ID and group ID)
- `useradd` (add user)
- `groupadd` (add group)
- `passwd` (change password)
- `chown` (change owner)
- `chmod` (change file permissions)



Basic Linux Commands

System Information

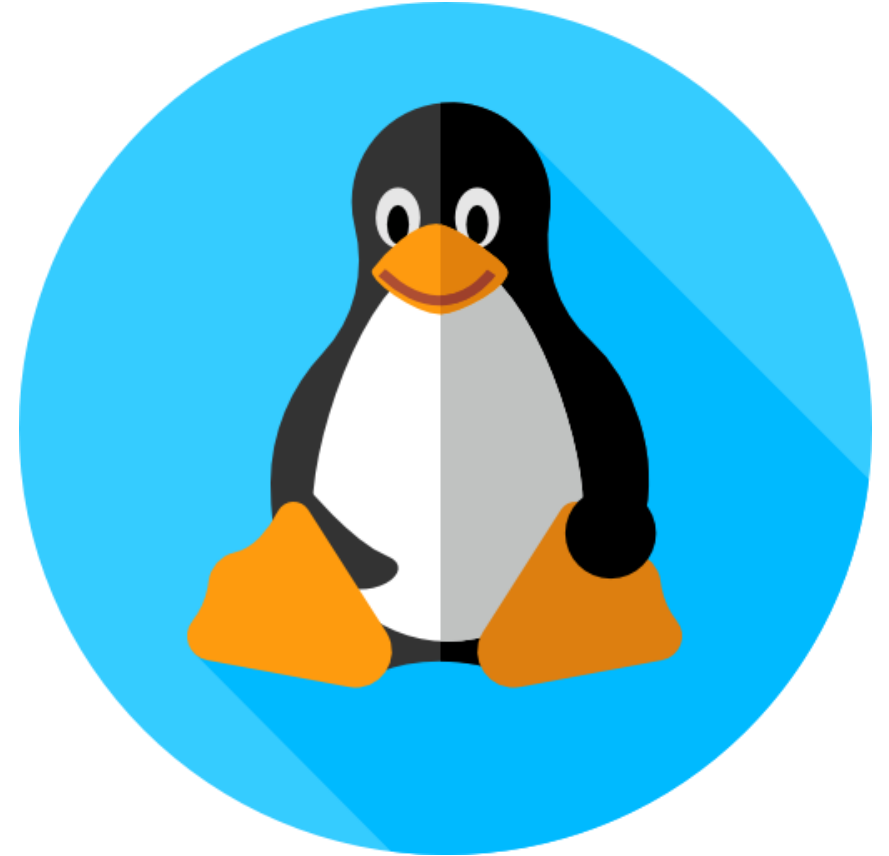
- `uname -a` (show system information)
- `df -h` (disk usage)
- `free -m` (memory usage)
- `top` (system monitor)
- `dmesg` (kernel ring buffer)



File Permissions and Ownership

Understanding Linux File Permissions

- Each file has three types of permissions:
 - **Read (r)**: View file contents
 - **Write (w)**: Modify file contents
 - **Execute (x)**: Execute file as a program
- **Permission Types:**
 - **Owner (u)**: The file creator
 - **Group (g)**: A group of users
 - **Others (o)**: Everyone else



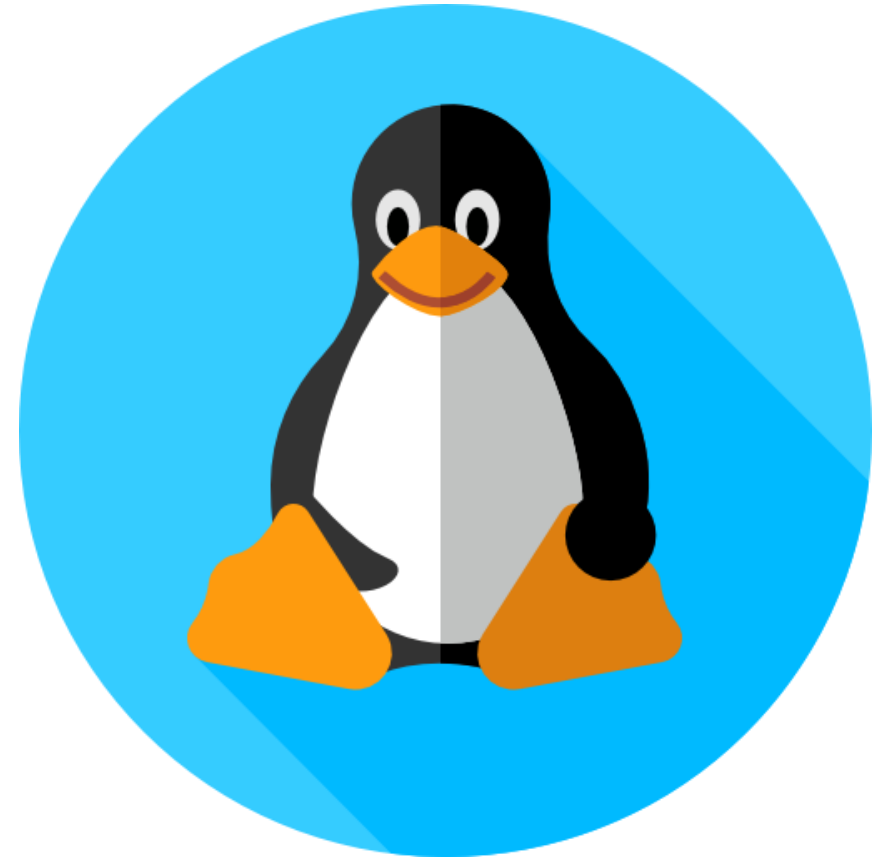
Package Management

Why Package Management?

- Package managers simplify software installation, updates, and removal.

Debian-based (Ubuntu)

- `apt` (Advanced Package Tool)
 - `apt update` (update package list)
 - `apt install package_name` (install a package)
 - `apt remove package_name` (remove a package)
 - `apt upgrade` (upgrade installed packages)



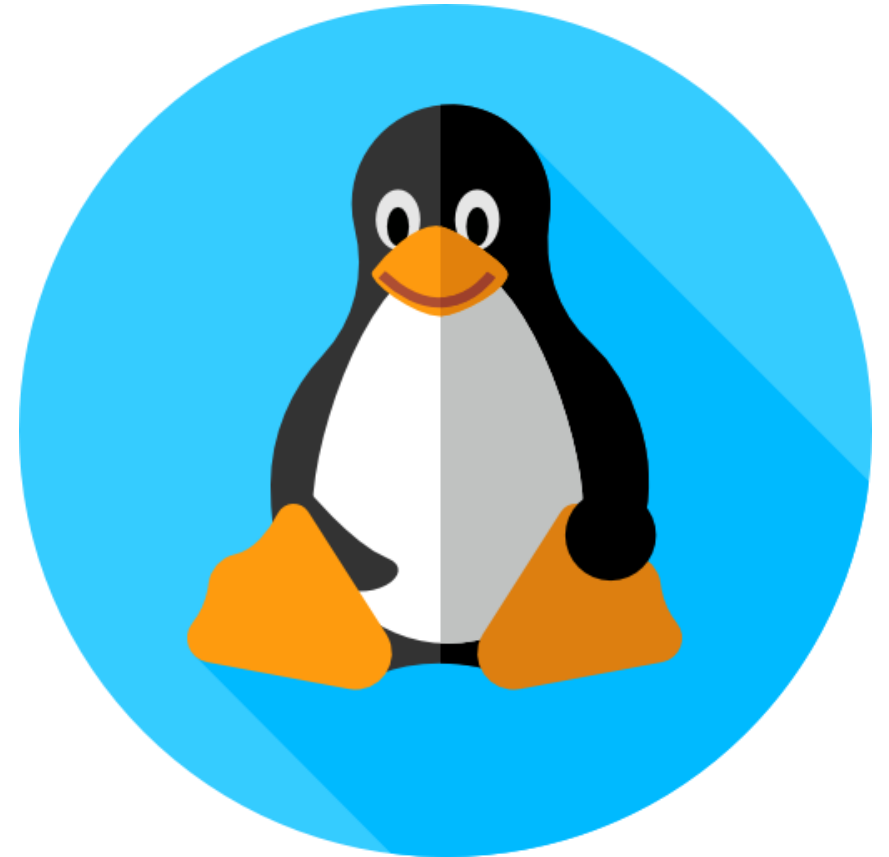
Package Management

Why Package Management?

- Package managers simplify software installation, updates, and removal.

Debian-based (Ubuntu)

- `apt` (Advanced Package Tool)
 - `apt update` (update package list)
 - `apt install package_name` (install a package)
 - `apt remove package_name` (remove a package)
 - `apt upgrade` (upgrade installed packages)



Networking and Services

Network Configuration

- `ifconfig` (show network interfaces)
- `ip a` (show IP address)
- `ping` (check connectivity)
- `curl` (retrieve data from URLs)



Networking and Services

Network Services

- **SSH (Secure Shell):** Remote login to other machines.
 - Command: `ssh user@hostname`
- **Firewall (iptables):** Control network traffic.
 - View active rules: `iptables -L`



Networking and Services

System Services (Systemd)

- `systemctl` (control system services)
 - `systemctl status service_name`
(check status)
 - `systemctl start service_name`
(start service)
 - `systemctl stop service_name`
(stop service)
 - `systemctl enable service_name`
(enable at boot)



Enough...

